SIH 2026 — PROBLEM STATEMENT SIH26002

RaahSetu: Live Demo Pitch & Judges Presentation Dialogues

Complete Step-by-Step Spoken Script in Natural Hinglish + English Mix for a Winning 5-Minute Live Web App Demonstration

PLATFORM URL	TARGET PITCH TIME	FORMAT	KEY TARGET
`http://127.0.0.1:8000/`	`4 to 5 Minutes`	`Action + Spoken Dialogue`	`Jury / Judges Panel`

⏱️ 5-Minute Pitch Structure at a Glance

Time	Demo Phase / Screen	Live Screen Action	Key Message to Convince Judges
0:00 - 0:45	1. The Life-or-Death Hook	Open `http://127.0.0.1:8000/` (Intro Page)	Why Google Maps fails in mountains (45.2% fatalities, ₹3500 Cr delays).
0:45 - 1:30	2. Driver Auth & Verification	Click "Sign In", solve Captcha, enter OTP	Bot-free security without third-party tracking cookies.
1:30 - 2:30	3. Vehicle & Cargo Manifest	**Step 2 Page:** Presets, GVW slider, Bailey bridge check	**The Game Changer:** Mountain bridge capacity & tunnel clearance enforcement.
2:30 - 3:30	4. Dual-Solve A* Engine	Route Dispatcher: Guwahati ➔ Shillong / Tawang	Fastest linear path vs. RaahSetu Risk-Aware corridor comparison.
3:30 - 4:15	5. 3D WebGL Terrain View	Click "3D Terrain View" button, pitch & orbit	Visualizing 14%+ steep slopes & cliff drop-offs before dispatch.
4:15 - 5:00	6. Multilingual Audio & Offline	Voice mic input + switch to Hindi/Assamese/Bengali	Hands-free driver safety & offline resilience in mountain dead zones.

🎤 Phase 1: The Strong Hook & Ground Reality (0:00 - 0:45 min)

`SCREEN ACTION` `Open browser at http://127.0.0.1:8000/. Stay on the Welcome/Intro page. Scroll slightly to show MoRTH accident numbers.`

"**Good morning respected Judges!**
Mera naam [Aapka Naam] hai, aur aaj hamari team introduce kar rahi hai **RaahSetu** — Bharat ke Northeast pahadi corridors ke liye pehla **Risk-Aware Logistics & Highway Accessibility Intelligence Platform**."

"Judges, plains me routing sirf ek cheez dekhti hai: *'Kahan jam kam hai aur kahan se jaldi pahuchein'*. Lekin Himalayas aur Northeast ghats me ek galat route lena **zindagi aur maut ka sawal** hota hai.
MoRTH (Ministry of Road Transport & Highways) aur NCRB ke official data ke mutabik, mountain roads par accident fatality rate **45.2%** hai — nearly double plain highways! Har saal 400 se zyada major landslides NH-10, NH-6 aur NH-13 jaise lifelines ko hafton tak band kar dete hain, jisse **₹3,500 Crore** ka freight delay hota hai aur oxygen cylinders, medical drugs aur food convoys raste me phas jate hain."

💡 **Judge Impact Tip:** Direct point bolna: *"Consumer navigation apps like Google Maps mountain physics ko samajhte hi nahi hain — wo multi-ton trucks ko 14% slope wale kacche shortcuts par bhej dete hain jahan brake fail ho jate hain."*

🎤 Phase 2: Driver Authentication & Bot-Free Security (0:45 - 1:30 min)

SCREEN ACTION `Top-right me "Sign In / Register" button par click karein. Auth page khulega. Canvas Captcha dikhayein aur "Verify OTP" karein.`

> "Ab hum aapko live platform ka demonstration dikhate hain.
> Jaise hi koi fleet driver, emergency dispatcher, ya state highway officer platform par aata hai, sabse pehle hum use karte hain ek **Two-Step Verified Authentication Flow**."

> "Judges, yahan dekhiye: Remote border checkposts par privacy maintain karne ke liye humne koi third-party tracking ya Google reCAPTCHA use nahi kiya. Humne ek **Procedural Client-Side Canvas Captcha** develop kiya hai jo character rotation aur noise dots ke sath brute-force automated bots ko 100% block karta hai.
> Driver apna credentials enter karta hai, email par 6-digit cryptographic OTP verify karta hai, aur instant access milta hai."

💡 **Judge Impact Tip:** "Verify" button dabayein — button par green tick aayega aur seedhe agla page open hoga!

🎤 Phase 3: The Game-Changer — Post-Login Vehicle & Cargo Manifest (1:30 - 2:30 min)

`SCREEN ACTION` `Login hote hi "Vehicle & Freight Specifications" (Step 2) page dikhega. 1-Click preset button dabayein, phir GVW slider ko move karein.`

> "**And Judges, THIS is where RaahSetu is radically different from any conventional navigation system!**
> Login karte hi system driver se seedhe destination nahi maangta — pehle wo vehicle ka **Physical & Structural Manifest** capture karta hai."

> "Yahan humne drivers ke liye **1-Click Quick Presets** diye hain:
> - 🚚 **Heavy 12-Wheeler Truck** (16.5 Tonnes, Multi-axle)
> - 💊 **Rapid Medical Carrier** (5.2 Tonnes, Oxygen & Critical Pharma)
> - 🛢️ **Petroleum POL Tanker** (14.2 Tonnes, Hazardous Fuel)
> - 🚙 **Mountain 4×4 Bolero Pickup** (High clearance)
>
> Ab Judges, dhyan se dekhiye is **Gross Vehicle Weight (GVW) Slider** ko! Pahar par Bailey Bridges aur suspension spans ki strict weight limits hoti hain:
> • Agar truck **12 Tonnes** se kam hai ➔ Real-time status shows: '🟢 *Cleared for all Northeast Bailey Bridges (Class 18/24)*'.
> • Agar truck **18 Tonnes** cross karta hai ➔ System alerts: '🔴 *Restricted: Re-routes away from sub-18T load-limited Bailey spans*'!"

> "Iske alawa Sonapur Tunnel (4.5m) aur Sela Tunnel (5.2m) ke liye height-width clearances check hote hain, aur driver Himalayan safety checklist (auxiliary retarder brakes aur snow chains) verify karta hai.
> Ab hum click karte hain: **'Confirm Vehicle & Plan Safe Route →'**."

> 💡 **Key Punchline:** *"Normal Google Maps ko pata hi nahi hota ki aap 16-ton truck chala rahe hain ya Maruti 800; RaahSetu har ek axle aur bridge ki bearing capacity ko route engine me feed karta hai."*

🎤 Phase 4: Route Dispatcher & Dual-Solve A* Engine (2:30 - 3:30 min)

`SCREEN ACTION` `Origin me Guwahati aur Destination me Shillong (ya Tawang) select karein. 2D Leaflet Map par dono routes appear honge. Screen par live telemetry badge dikhayein.`

> "Ab hum aa gaye hain core **Route Dispatcher Workspace** par.
> Screen par dekhiye: Top par live telemetry badge dikh raha hai — **'• FastAPI A* Connected (XXms)'**. Yeh hamara custom-built Python FastAPI graph engine hai jo sub-millisecond me routes evaluate kar raha hai."

> "Yahan screen par do alag-alag routes calculate huye hain — ise hum kehte hain **Dual-Solve Algorithm**:
> 1. **Fastest Route (Conventional Nav):** Ye route sirf linear distance (km) kam karta hai, lekin isme steep elevation gradients aur landslide risk points aate hain.
> 2. **RaahSetu Risk-Aware Recommended Corridor (Green Glowing Polyline):** Ye route dhalan (slope), bridge load rating, aur weather advisories ko optimize karke banaya gaya hai. Chahe 15-20 minute extra lag jayein, lekin brake fade aur rollover ka risk 60% se zyada drop ho jata hai."

> 💡 **Technical Rationale:** Hamara objective cost function hai: `Cost = TravelTime + λ × Risk_Exposure + Bridge_Penalty + Slope_Penalty`.

🎤 Phase 5: 3D WebGL Terrain Elevation Inspection (3:30 - 4:15 min)

SCREEN ACTION `Map ke top-right par "3D Terrain View" toggle button click karein. Three.js canvas render hoga. Mouse se tilt, pitch aur zoom karke mountain passes dikhayein.`

> "**Now Judges, let us show you something that no other logistics software has:**
> Hum click karte hain **'3D Terrain View'** par.
> Ye hamara custom **Three.js aur React Three Fiber (WebGL)** based 3D elevation engine hai!"

> "Judges, 2D flat map par pahad ki dhalan nahi dikhti. Yahan 3D me driver aur fleet dispatcher actual pahadon ki unchai dekh sakte hain:
> • Brahmaputra river valley lowland green color me hai.
> • Foothills olive brown me hain.
> • Aur high Himalayan passes (jaise Sela Pass, 13,700 feet) snow-white peaks me render ho rahe hain.
> Dispatcher ise rotate karke dekh sakta hai ki raste me ghat hairpin bends aur vertical drop-offs kahan hain, taaki heavily loaded trucks ko unsafe inclines par bhejne se pehle hi alert kiya ja sake."

💡 **Wow Factor:** 3D scene ko smoothly rotate karein. Judges visual elevation mesh dekh kar turant impress hote hain!

🎤 Phase 6: Multilingual Voice Audio & Mountain Offline Resilience (4:15 - 5:00 min)

SCREEN ACTION `Search box me Mic icon click karein aur bole: "Shillong". Phir header me Language dropdown se Hindi, Assamese, ya Bengali choose karein.`

> "Pahadi rasto par truck chalate waqt driver mobile par type nahi kar sakta. Isliye humne integrate kiya hai **Hands-Free Web Speech Recognition**:
> Driver sirf bolta hai *'Shillong'* ya *'Guwahati'*, aur hamara phonetic fuzzy matching engine regional accent ko auto-correct karke location lock kar leta hai."

> "Iske alawa humne Bharat ke Northeast ke liye **Full Multilingual Localization** implement kiya hai — English, Hindi, Assamese (অসমীয়া), aur Bengali (বাংলা). Pure road warnings, city names aur audio turn-by-turn guidance driver ko uski matribhasha me sunayi deti hai."

> "**And what if internet completely dies in remote passes?**
> RaahSetu ka frontend **100% Offline Resilient** hai. Humne 111 cities aur 378 road corridors ka graph browser me pre-cache kiya hai. Agar server se connection tut bhi jaye, tab bhi frontend ka in-browser Dijkstra solver bina internet ke safest route calculate karta hai, aur emergency SOS alerts LocalStorage me queue hokar signal aate hi auto-flush ho jate hain."

> "**Conclusion:** RaahSetu sirf ek software nahi hai, ye mountain logistics ke har truck driver aur strategic corridor ke liye ek **suraksha kavach** hai. Thank you, and we are now open for your questions!"

🎯 Top 5 Judge Cross-Questions & Killer Answers (Q&A Cheat Sheet)

Q1: "Google Maps already exists and has huge data, why would anyone use RaahSetu?"

Answer: "Sir, Google Maps consumer passenger cars ke liye optimize karta hai — wo dhalan (gradient), brake overheating, multi-axle bridge weight limits (Bailey Class 18/24), aur hazardous tanker sensitivity ko bilkul ignore karta hai. RaahSetu ek specialized *Heavy Freight & Strategic Corridor Engine* hai jo vehicle gross weight, tunnel portal height, aur MoRTH hazard data ko route calculation ka primary constraint banata hai."

Q2: "What is your tech stack and how is this deployed?"

Answer: "Frontend React 19 + TypeScript + Vite 7 + Tailwind CSS v4 + Three.js + Leaflet par bana hai. Backend Python FastAPI + Pydantic + PostGIS graph engine hai. Sabse khaas baat ye hai ki humne **Single-Port Unified Delivery** implement kiya hai: FastAPI backend root par hi frontend dist bundle serve karta hai (port 8000), jisse zero CORS issues aate hain aur single link par complete application chal rahi hai."

Q3: "Agar pahad me internet chala gaya toh system kaise chalega?"

Answer: "Sir, humne Offline-First Architecture design kiya hai. 111 surveyed nodes aur 378 road corridors ka verified graph client-side JavaScript me embedded hai. Internet disconnect hone par client-side Dijkstra solver 2 milliseconds me rasta nikalta hai, aur offline hazard reports LocalStorage queue me save hokar signal restore hote hi transmit ho jaati hain."

Q4: "Data kahan se validate hua hai?"

Answer: "Humne official government datasets use kiye hain: MoRTH Road Accident Census, Geological Survey of India (GSI) landslide databases, OpenStreetMap real road network geometries, aur BRO (Border Roads Organisation) Bailey bridge load classifications."

Q5: "Can this be scaled across India?"

Answer: "Absolutely Sir! Hamara graph snapshot model modular hai. Jaisa graph humne Northeast ke 8 states ke liye banaya hai, waisa hi snapshot Western Ghats (Maharashtra, Kerala), Himachal Pradesh, Uttarakhand, aur Jammu-Kashmir ke liye seamlessly plug kiya ja sakta hai."